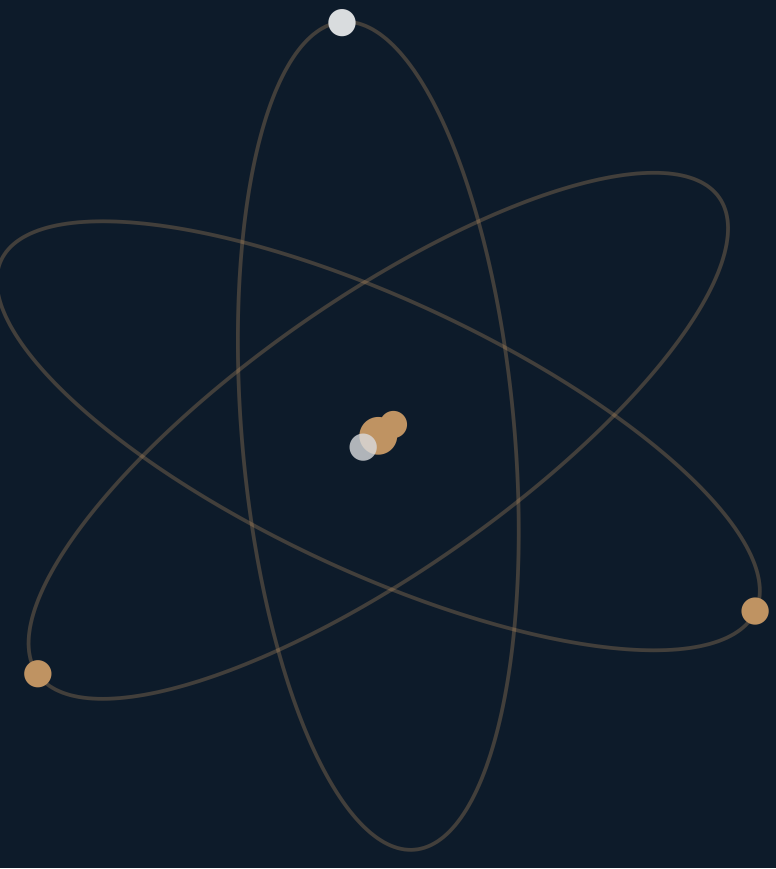
APPLIED PHYSICS

Mid-Term Study Guide



Contents

Foundations: science, theory and the scientific method 5

Key definitions 5
Scope of a theory 6
Logical reasoning in the scientific method 6
The scientific method 7
Types of variables 8
Graph of independent against dependent variable 8
Physical quantities 9

Unit 1: Vectors, relativity and physical theories 13

1. Vectors and physical quantities 13
2. Principles of relativity 17
3. Theoretical and applied physics 19
4. Major physical theories and their applications 21
5. Quantum mechanics: three revolutions 22
6. Applications of physics in Artificial Intelligence 23

Unit 2: Electric charge, quarks and fields 25

7. Electric charge and fundamental charges 25
8. Fundamental structure of nucleons 26
9. Fields and their fundamental nature 27
10. Types and sources of fields 28

Unit 3: Electrostatics 29

11. Coulomb’s law 29
12. Electric field and electric field intensity 31
13. Continuous distribution of electric charge 33

14. Electric flux 35
15. Electric dipole 36

Lab activities (not included in the mid-term) 38

Lab Activity 1: Ohm’s law 38
Lab Activity 2: errors and least count 39
Vernier calliper and screw gauge 40

Quick revision 42

Formulas to write from memory 42
Common mistakes 42
Self-test 43

The Exam Map

What each unit covers.

Unit	Topics
Foundations	Science, theory, law, scientific method, variables, physical quantities
1. Vectors, relativity, theories	1 Vectors and invariants 2 Galilean and special relativity 3 Theoretical vs applied physics 4 Major theories 5 Three quantum revolutions 6 Physics in AI
2. Charge, quarks, fields	7 Charge types 8 Nucleon structure 9 Fields 10 Types and sources of fields
3. Electrostatics	11 Coulomb’s law 12 Electric field 13 Continuous charge 14 Electric flux 15 Dipole
Labs	Activity 1 (Ohm’s law), Activity 2 (least count, errors), Vernier and screw gauge

HOW TO USE THIS GUIDE

The lab activities are marked **not included in the mid-term** in your notes, so they sit in their own section near the end. Topics marked “syllabus only” in your notes are standard physics: learn the tables, not long derivations.

Foundations: science, theory and the scientific method

Key definitions

Term	Definition
Science	A testable qualitative or quantitative statement (hypothesis, assumption, law, model or theory) that can explain a natural phenomenon through repeated observations and experiments.
Experiment	A systematic procedure to investigate a natural phenomenon and establish a **cause-and-effect** relationship between independent variables, using devices in labs under **controlled conditions** (e.g. temperature and pressure).
Theory	A testable framework made of a set of **assumptions or postulates** that explains a phenomenon in terms of cause and effect, that is, the relationship between independent variables.
Law	A qualitative formalism or statement based on a **large number of experimental observations**.

Example experiment: investigate current electricity in a conductor and establish the relationship between current and voltage.

KEY IDEA

When there is no direct experimental evidence to confirm a theory, it can be tested indirectly through **hypothetical (thought) experiments**, based on biological, mathematical and philosophical reasoning.

Scope of a theory

1. Explains natural phenomena.
2. Explains *why* some phenomena occur.
3. Explains the cause-and-effect relationship between dependent and independent variables.
4. Explains the meaning of experimental data.
5. **Predicts** something new about a natural phenomenon.
6. **Guides experimentalists** to investigate natural phenomena through observations and experiments.
7. Is **broader in scope**, explaining a larger set of experimental data and observations.

Logical reasoning in the scientific method

- Logical reasoning is the process of using **facts, evidence and clear thinking** to reach a correct conclusion.
- It is important in rational arguments in daily life, in scientific experiments and in the formation of theories.
- A **rational argument** is a statement in which reasons, facts, evidence and logical principles support a conclusion, for example the results of an experiment.
- It has two parts: **premises** (base statements, evidence, observations) and a **conclusion** (a judgement derived from them).

LECTURE EXAMPLE

Premise 1: metal A expands on heating.

Premise 2: metal B expands on heating.

Conclusion: all metals expand on heating.

The premises are specific observations and the conclusion is a general law.

The scientific method

A systematic investigation of a natural phenomenon. Steps, with the Ohm's law example from class:

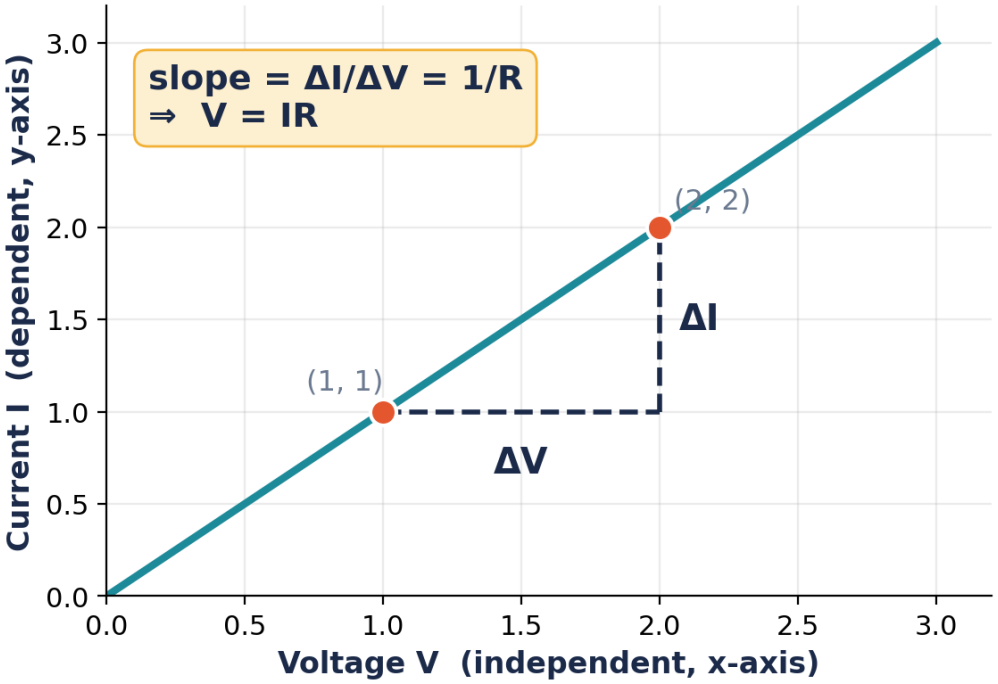


Types of variables

Variable	Meaning	Example (Ohm's experiment)
Independent	The variable we change on our own; it does not depend on the other variable in that experiment	Voltage V
Dependent	Depends on the other variables and changes when the independent variable changes	Current I
Constant	A quantity that stays the same in the experiment	
Controlled	A variable held fixed externally	Temperature, resistance or length of the conductor

Graph of independent against dependent variable

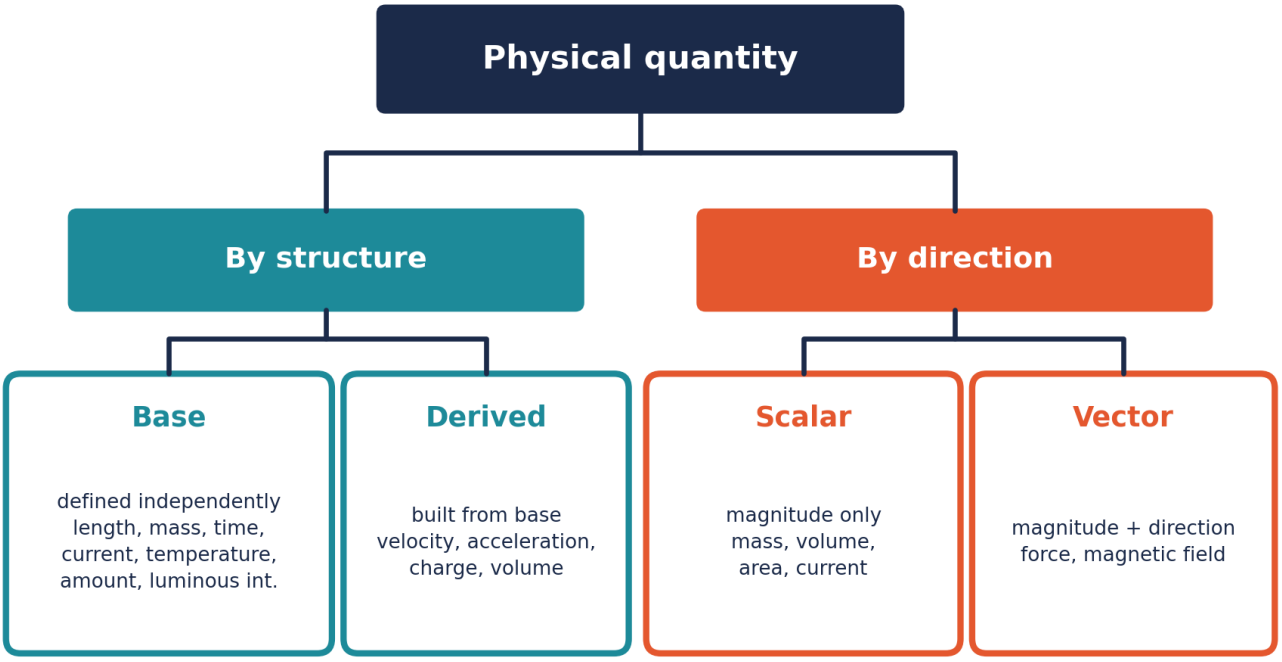
- Convention: the **independent** variable goes on the horizontal (**x**) axis and the **dependent** variable on the vertical (**y**) axis. In Ohm's experiment that means V along x and I along y.
- Sample data from class: V = 1, I = 1, R = 1 and V = 2, I = 2, R = 1. R stays constant, so the plot is a **straight line through the origin**.
- Slope = $\tan\theta = \Delta I / \Delta V = 1/R$, which gives $\Delta V = \Delta I \cdot R$, the form of Ohm's law.



slope = ΔI / ΔV = 1 / R ⇒ V = I R

Physical quantities

- A physical quantity can be **measured accurately with an instrument**, for example length with a meter scale or mass with a balance. Quantities such as happiness cannot be measured with a device and are called un-physical quantities.
- Why they matter: they are measurable, they describe the properties of an object, and they are used in laws, theories and models. We test a hypothesis, law, model or theory by measuring physical quantities.



By structure

Type	Meaning	Examples
Base (fundamental)	A standard defined for one aspect of an object, defined independently, so it cannot be derived from other quantities	Length, mass, time, current, temperature, amount of substance, luminous intensity
Derived	Defined in terms of base quantities	Volume ($4/3\ \pi r^3$ for a sphere, L^3 for a cube), velocity, acceleration, charge

The seven SI base quantities

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Dimensions of derived quantities

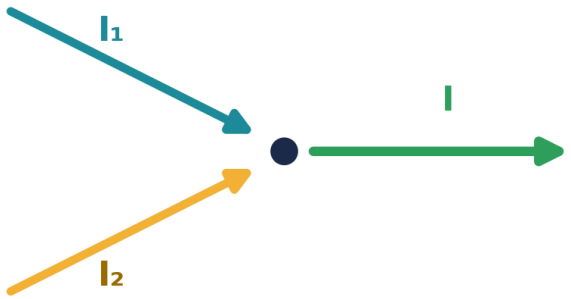
Quantity	Formula	Dimension
Velocity	$v = dl/dt$	LT^{-1}
Acceleration	$a = d^2l/dt^2$	LT^{-2}
Charge	$Q = It$ (from $dQ/dt = I$, so $dQ = I\,dt$, then integrate)	A·s

By direction

- **Scalar:** magnitude only. Examples: mass, volume, area, current.
- **Vector:** magnitude and direction, **and it obeys vector addition**. Examples: force, magnetic field.

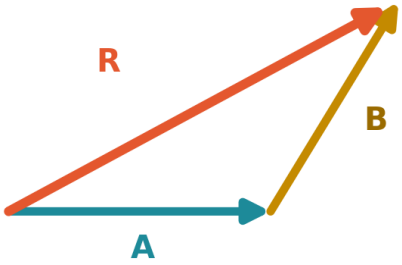
Why is current a scalar even though it has a direction?

Currents at a junction



I = I₁ + I₂ (plain sum)

Vectors (triangle law)



|R| ≠ |A| + |B|

WATCH OUT

Current flows along the wire, but it does **not** obey the vector (triangle) law. If two currents I_1 and I_2 meet at a junction, the total is the plain sum $\mathbf{I = I_1 + I_2}$, not the vector sum of two arrows at an angle. A quantity that does not add like a vector is not a vector.

Unit 1: Vectors, relativity and physical theories

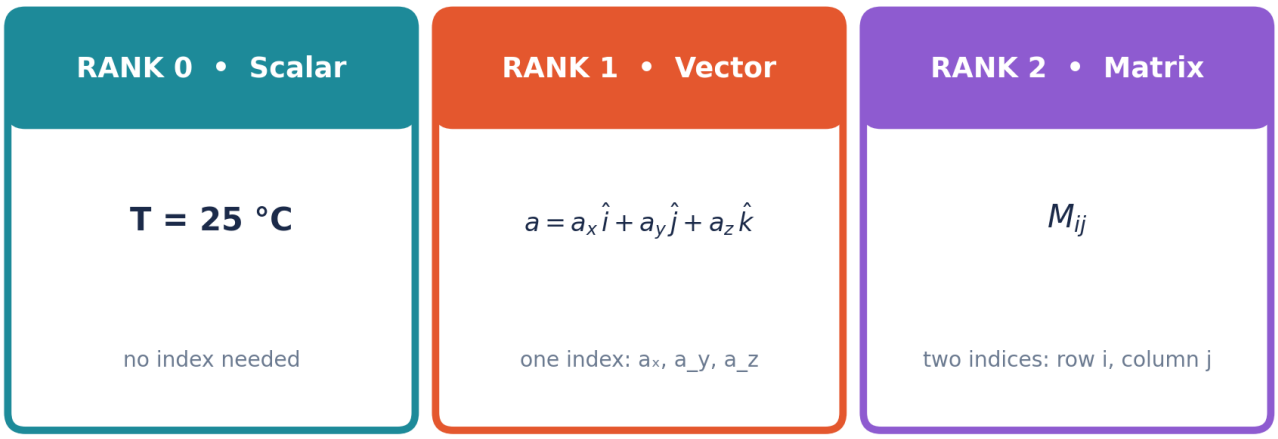
1. Vectors and physical quantities

SIMPLE DEFINITION

A **vector** is a quantity with both magnitude and direction that adds by the vector (triangle) law. A **scalar** has magnitude only.

What “tensor of rank one” means

Rank is the number of **indices** needed to label a quantity’s components. So a vector is a tensor of rank one.



Vector space properties

A vector space is a set of vectors where adding them and multiplying them by numbers always gives another vector in the set. Its elements obey these laws of addition and multiplication:

1. **Closure:** adding two elements, or multiplying one by a scalar, gives another element of the space.

2. **Associative** law of addition and multiplication.
3. **Commutative** law of addition and multiplication.
4. **Inverse** element.
5. **Identity** element.

> **EXAM TIP**
> The two laws most often misspelled are the **closure** property and the **commutative** law.

Show by closure that a 2×2 matrix is a vector

Add two 2×2 matrices entry by entry and you get another 2×2 matrix, so the set is closed under addition. Multiplying by a scalar k multiplies every entry by k, giving another 2×2 matrix (closed under scalar multiplication).

$$M_1 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad M_2 = \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}$$

$$M_1 + M_2 = \begin{bmatrix} a+a' & b+b' \\ c+c' & d+d' \end{bmatrix} \qquad k\,M_1 = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

The other vector-space laws also hold for matrix addition, so the 2×2 matrices form a vector space and each matrix is a vector.

Show that a combination of e^{ikx} and e^{-ikx} is a vector

$$\psi = A\, e^{ikx} + B\, e^{-ikx}$$

Closure under addition: add two such combinations and collect terms. The result has the same form with new constants.

$$\psi_1 + \psi_2 = (A_1 + A_2)\, e^{ikx} + (B_1 + B_2)\, e^{-ikx}$$

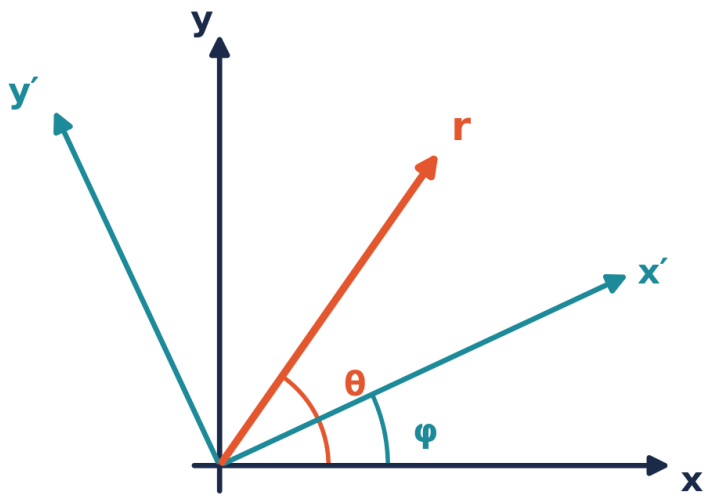
Closure under scalar multiplication: multiply by a constant c and the result again has the same form.

$$c\,\psi = (cA)\, e^{ikx} + (cB)\, e^{-ikx}$$

Closure holds, so these combinations form a vector space and ψ is a vector.

Show that the position vector is invariant under rotation of coordinates

In 2D the position vector is $\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j}$ with $x = r\cos\theta$, $y = r\sin\theta$, and magnitude $|r| = \sqrt{x^2 + y^2}$. Rotate the axes by an angle φ:



$|r'| = |r|$

$$x' = x\cos\varphi + y\sin\varphi,\ y' = y\cos\varphi - x\sin\varphi$$

Square and add. The cross terms **2xy sin φ cos φ cancel** and $\cos^2\varphi + \sin^2\varphi = 1$, so:

$$|r'| = \sqrt{(x'^2 + y'^2)} = \sqrt{(x^2 + y^2)} = |r|$$

The magnitude is unchanged, so the position vector is **invariant**.

EXAM TIP

Your slide writes $y' = r\sin\varphi' = r\sin(\theta' - \varphi)$. The working value is $y' = r\sin(\theta - \varphi) = r\sin\theta\cos\varphi - r\cos\theta\sin\varphi = y\cos\varphi - x\sin\varphi$.

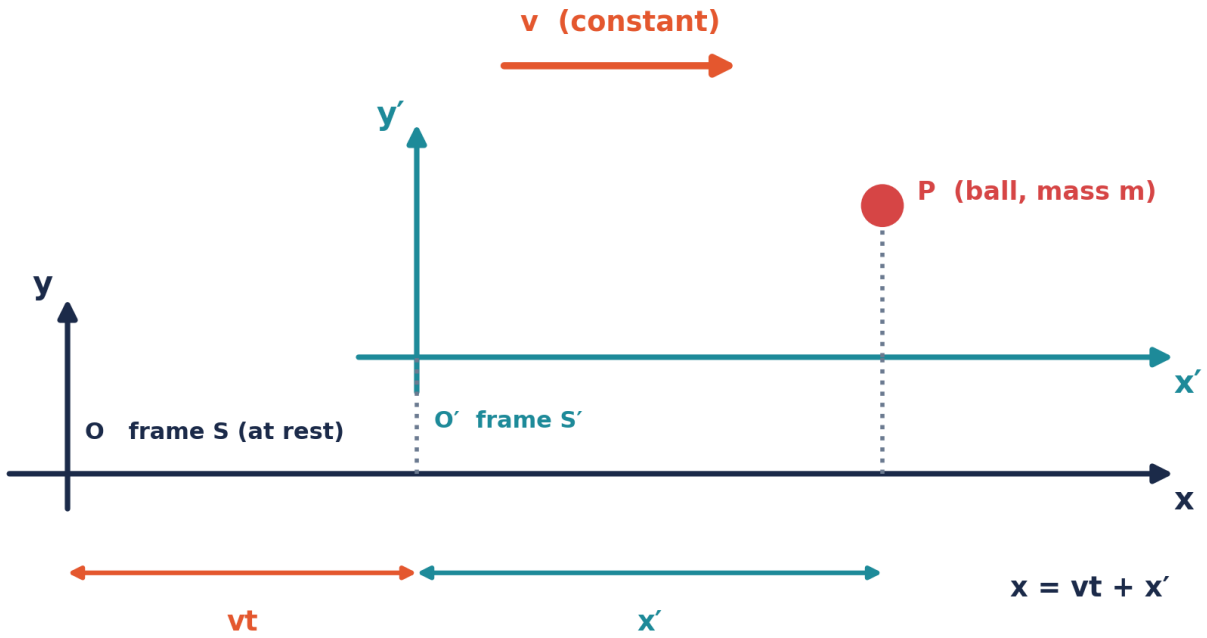
2. Principles of relativity

GALILEAN RELATIVITY

Principle of Galilean (Newtonian) relativity: all laws of mechanics remain the same in all inertial frames of reference. An **inertial frame** is one that is at rest or moves with constant velocity (no acceleration).

Invariance of Newton's law in inertial frames

Setup: two inertial frames, S (x, y, z, t) at rest and S′ (x′, y′, z′, t′) moving with constant velocity **v** along the x-axis. Both coincide at the origin when t = 0. After time t, S′ has moved a distance vt. A ball of mass m sits at point P, with velocity u′ relative to S′ and u relative to S.



Step 1. Position of the ball relative to S:

$$x = vt + x' \text{ or } x' = x - vt$$

The equations have the same form in both frames, so the physics is the same for observers in S and S′. This concept is called **invariance**.

Step 2. Differentiate with respect to time to get the velocity addition law (Galilean transformation):

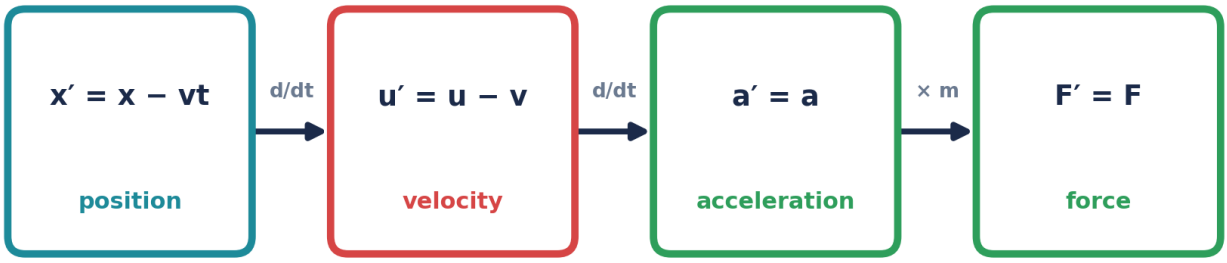
$$dx/dt = v + dx'/dt \Rightarrow u = v + u' \text{ or } u' = u - v$$

Step 3. Differentiate again. Since v is constant, dv/dt = 0:

$$du/dt = du'/dt \Rightarrow a = a'$$

Step 4. The mass is the same in both frames, so m a = m a′:

$$F = m\,a = m\,a' = F'$$



Velocity CHANGES between frames (u ≠ u′). (v is constant, so dv/dt = 0.)

Acceleration and force are INVARIANT, so Newton's laws keep the same form.

REMEMBER

Newton's laws have the same form (are invariant) in both S and S′. **Acceleration and force are the same** for both observers, while **velocity is not** (u ≠ u′).

Quick check from the board sketch: two velocities in the same direction subtract ($V = v_1 - v_2$), and in opposite directions they add ($V = v_1 + v_2$).

Principles of special relativity

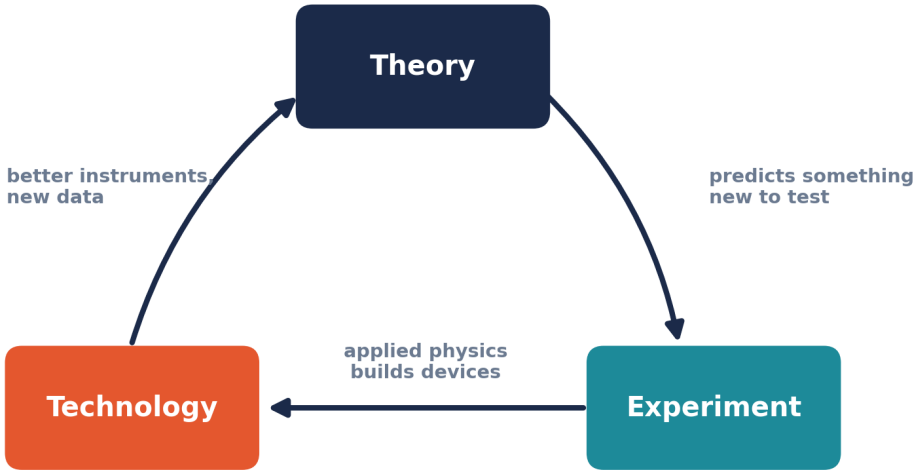
- **Postulate 1:** the laws of physics are the same in all inertial frames.
- **Postulate 2:** the speed of light in vacuum, **c**, is the same for all inertial observers, regardless of the motion of the source or the observer.
- Galilean addition $u = v + u'$ fails near the speed of light. The correct transformations are the **Lorentz transformations**, which mix space and time, giving **time dilation** and **length contraction**.
- Galilean relativity is the **low-speed limit** (v much less than c) of special relativity.
- **Maxwell's equations** are not invariant under Galilean transformations, but are invariant under Lorentz transformations. This was a main motivation for special relativity.

$$\mathbf{x' = \gamma\,(x - vt),\ t' = \gamma\,(t - vx/c^2),\ \gamma = 1 / \sqrt{(1 - v^2/c^2)}}$$

3. Theoretical and applied physics

Syllabus only for this topic, so check it against your class notes.

	Theoretical physics	Applied physics
Aim	Understand nature: build models, theories and mathematical frameworks that explain and predict phenomena	Use known physics to solve practical problems and build technology
Method	Mathematics, principles, thought experiments, predictions	Experiments, devices, engineering, measurement
Output	Laws, theories, equations	Technologies, instruments, materials, products
Example	Bohr’s atomic theory, energy band theory	Lasers, semiconductors, sensors, detectors



The two feed each other: theory predicts, experiments test, and new technology provides better experiments.

Basic criteria of a physical theory

It must be **testable**, explain natural phenomena and the meaning of experimental data, explain cause and effect, **predict something new**, guide experimentalists, and have broad scope.

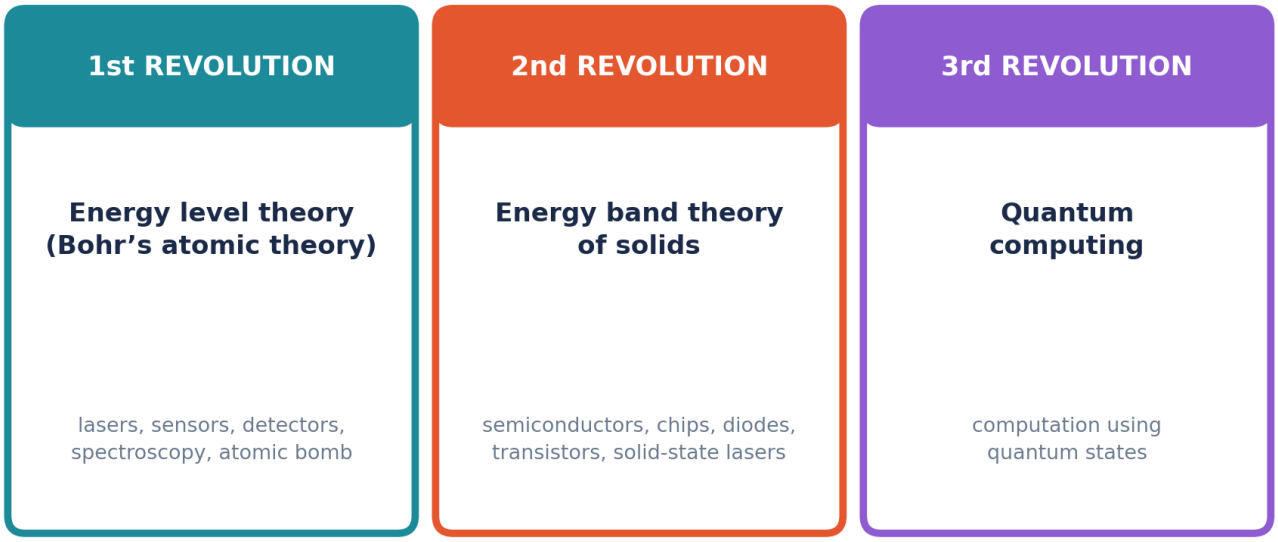
General applications of physical theories

Explaining natural phenomena, predicting new ones, designing devices and materials, guiding experiments, and building technologies such as computers, medical imaging and communication.

4. Major physical theories and their applications

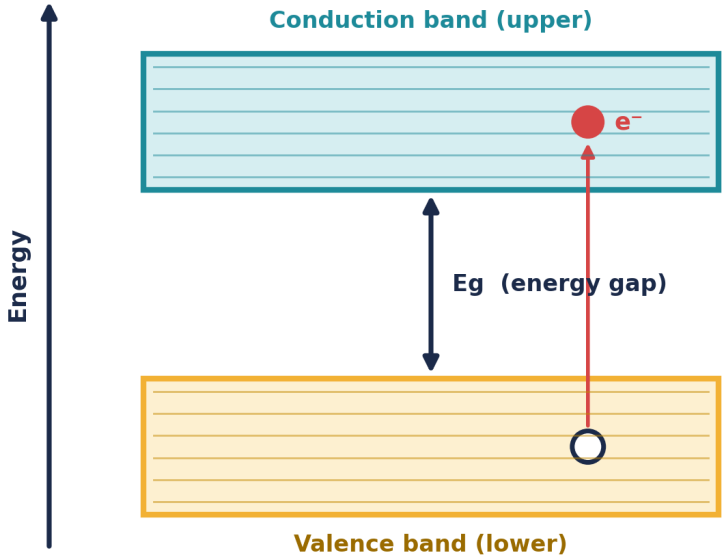
Theory	What it describes	Typical applications
Classical mechanics	Motion of everyday-scale objects under forces (Newton’s laws)	Vehicles, machines, structures, robotics, trajectories
Thermodynamics	Heat, work, energy and entropy in bulk matter	Engines, refrigerators, power plants, energy efficiency
Electromagnetic theory	Electric and magnetic fields and light (Maxwell)	Motors, generators, communications, optics, electronics
Statistical mechanics	Bulk behaviour from the statistics of many particles	Gases, phase transitions, materials, probability models
Quantum mechanics	Atoms, electrons and light at small scales	Lasers, semiconductors, spectroscopy, quantum computing

5. Quantum mechanics: three revolutions



No.	Theory	What it explains	Applications
First	Energy level theory / atomic theory (Bohr)	Structure, formation, properties (binding energy, energy levels) and interactions of atoms	Atomic bomb, sensors, detectors, laser, spectroscopy, nanotechnology, quantum computing
Second	Energy band theory of solids	When we group electrons they form energy bands. A conduction band sits above a valence band, separated by a gap Eg	Semiconductors (nanotechnology), computer and iPhone chips (silicon), semiconductor lasers, solid-state detectors, diodes, rectifiers, transistors, amplifiers
Third	Quantum computing	Computation using quantum states	Quantum computing

Band picture from the notebook



The valence band is the lower band, the conduction band is the upper band, and **Eg** is the energy gap between them. Within each band the energy levels are continuous.

6. Applications of physics in Artificial Intelligence

Syllabus only for the table below.

Physical theory	Application in AI
Classical mechanics	Robotics and motion planning, self-driving vehicles, physics engines for simulation and reinforcement learning, gradient descent viewed as a ball rolling downhill
Thermodynamics	Energy use and heat limits of AI hardware and data centres, cooling, simulated annealing, entropy in information theory
Electromagnetism	Electronics, chips, memory, sensors, cameras, wireless communication
Statistical mechanics	Boltzmann machines, probabilistic models, Monte Carlo methods, energy-based models, neural network statistics
Quantum mechanics	Semiconductor chips, quantum computing and quantum machine learning

Unit 2: Electric charge, quarks and fields

7. Electric charge and fundamental charges

- Charge is an **intrinsic property** of any material object, due to which it interacts with other objects.
- More fundamentally, charge is a **quantum number**, defined to explain the properties and interactions of matter observed in various experiments.

Types of charge

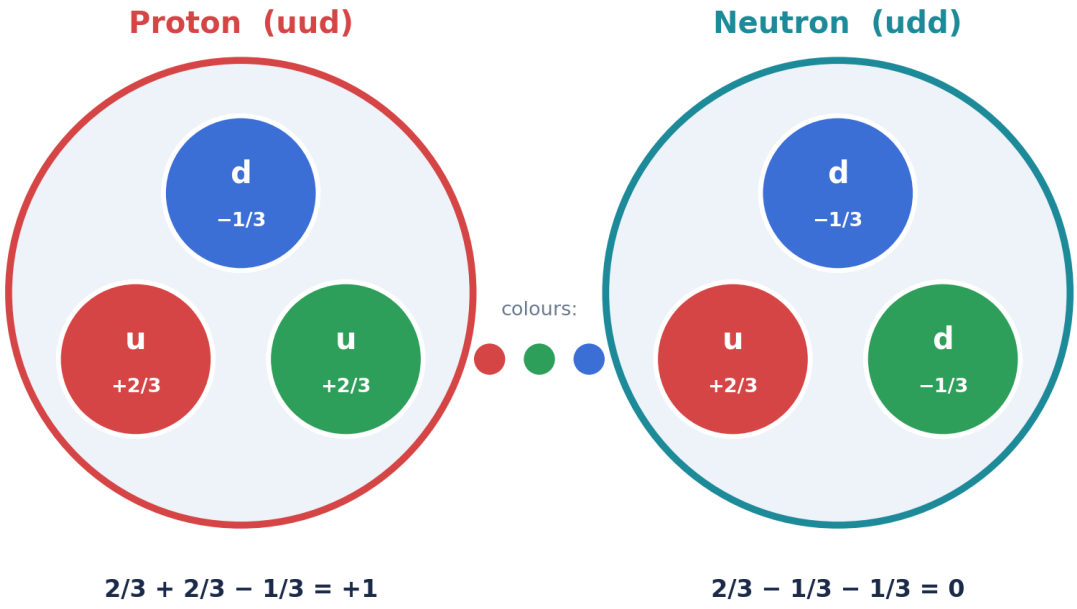


#	Type	Interaction	Notes
1	Electric charge	Electric forces and electromagnetic fields	
2	Gravitational charge (mass)	Gravity	$F = G\, m_1 m_2 / r^2$
3	Weak charge	Weak nuclear force	Carried by $W^{\pm}$ and Z^0 (weak bosons); example: beta decay
4	Strong nuclear charge (colour charge)	Strong force	Three colours: red, green, blue; example: colour charge on quarks

8. Fundamental structure of nucleons

Syllabus only for this topic.

- Nucleons are **protons and neutrons**. Each is made of **three quarks**.
- Quarks have fractional electric charge: **up quark u = +2/3 e**, **down quark d = −1/3 e**.
- **Proton = uud.** Charge = 2/3 + 2/3 − 1/3 = **+1**.
- **Neutron = udd.** Charge = 2/3 − 1/3 − 1/3 = **0**.
- Quarks have spin 1/2 and carry colour charge (red, green, blue). A nucleon is colourless overall, with one quark of each colour.



PAULI AND COLOUR

Pauli exclusion principle: no two identical fermions (spin-1/2 particles such as quarks and electrons) can occupy the same quantum state. In a proton the two u quarks are identical, so they must differ in another quantum number, which is why quarks have **colour**.

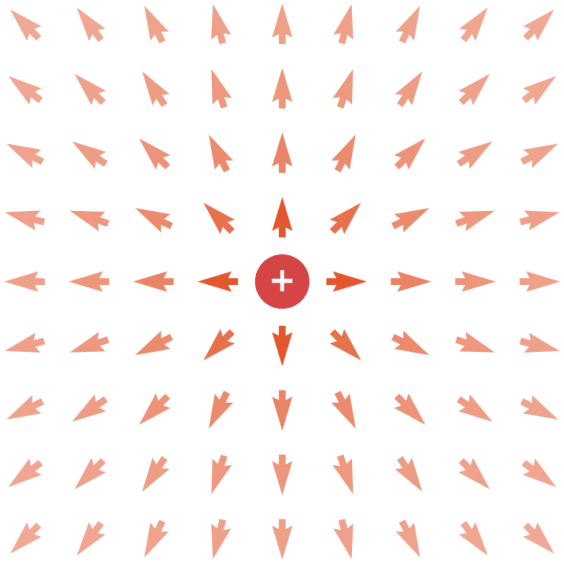
9. Fields and their fundamental nature

- A **field** is a region of space in which every point is represented by a physical quantity, which can be a scalar or a vector.
- **Scalar field:** a number at every point. Example: temperature field.
- **Vector field:** a vector at every point. Examples: electric, magnetic and gravitational fields.
- Fields are **fundamental** entities in modern physics. Particles are quanta of their fields: the **photon** is a quantum of the electromagnetic field and the **electron** is a quantum of the electron field.

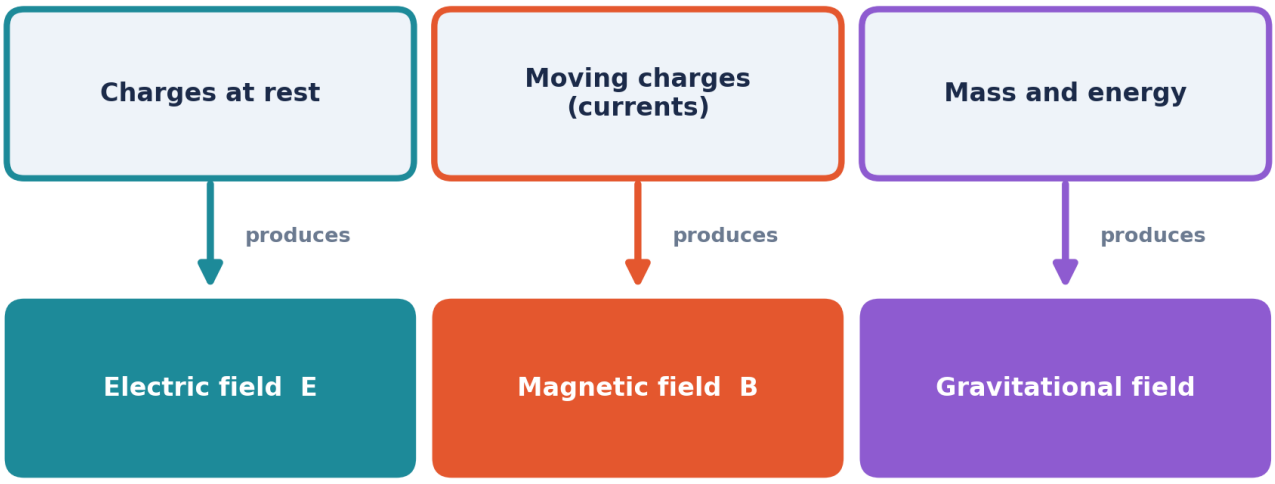
Scalar field: temperature (one number at each point)

18°	20°	23°	26°	29°
19°	22°	25°	29°	32°
20°	24°	28°	32°	35°
19°	22°	26°	30°	33°
18°	20°	23°	27°	30°

Vector field: electric field of +Q (an arrow at each point)



10. Types and sources of fields



Field	Produced by
Electric field	Electric charges; charges at rest can produce an electric field
Magnetic field	Moving electric charges (electric currents)
Gravitational field	Mass and energy

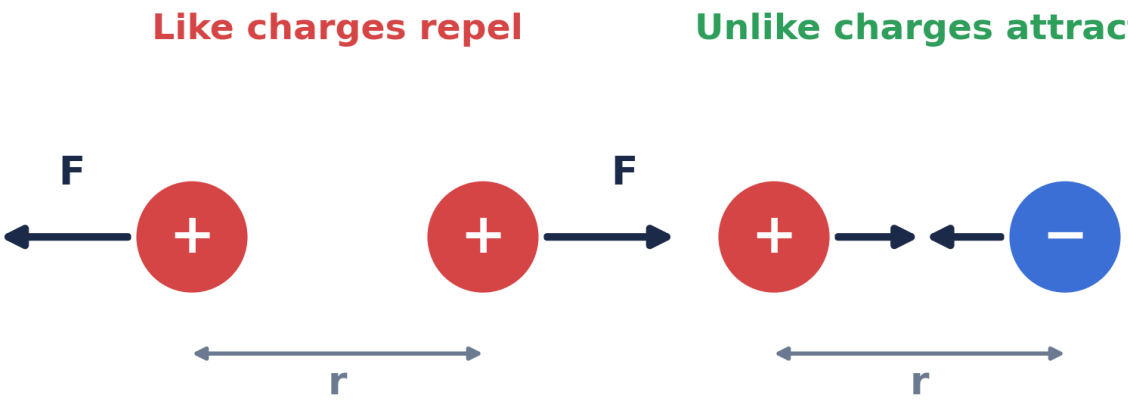
Unit 3: Electrostatics

Electrostatics is the branch of physics that deals with the study of electric charges at rest.

11. Coulomb’s law

COULOMB’S LAW

The force between two stationary point charges is **proportional to the product of the charges** and **inversely proportional to the square of the distance** between them. It acts along the line joining them: **repulsive** for like charges, **attractive** for unlike.



$$F = k\, q_1 q_2 / r^2$$

$$F = k\, q_1 q_2 / r^2,\ k = 1/(4\pi\, \varepsilon_0) \approx 9 \times 10^9\ N\ m^2/C^2$$

Vector form: $F = k\, q_1\, q_2\, \hat{\mathbf{r}} / r^2$, where $\hat{r}$ is the unit vector along the line joining the charges.

Worked example

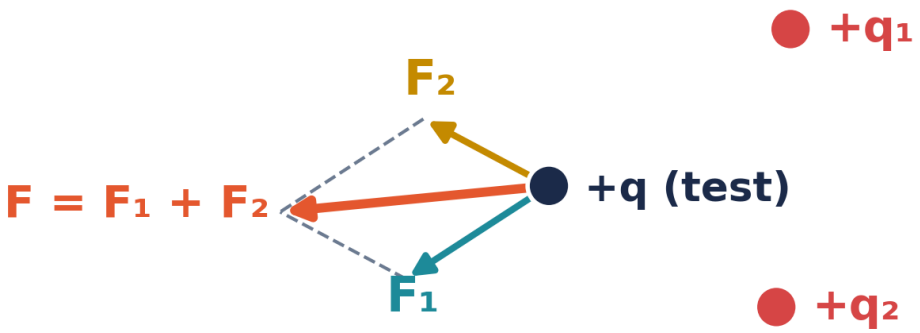
q_1 = 2 μC and q_2 = 3 μC at r = 10 cm.

```
F = (9×10⁹)(2×10⁻⁶)(3×10⁻⁶) / (0.1)²
F = 5.4 N   (repulsive, both positive)
```

Superposition of forces and fields

The force between two charges (for example a test charge and a source charge) is **unaffected by the presence of other charges**. To find the force on a test charge +q from source charges $+q_1$, $+q_2$, …, $+q_n$:

1. Calculate F_1 on +q due to q_1, ignoring all other charges.
2. Repeat to get F_2, F_3, …, F_n, taking one at a time.
3. The total force is the **vector sum**.



$$F = F_1 + F_2 + \ldots + F_n = \Sigma F_i = k\, q\, \Sigma \left(q_i \hat{r}_i / r_i^2 \right)$$

This is the **principle of superposition**.

Limitations of Coulomb’s law

#	Limitation
i	Applies only to **stationary** charges
ii	Applies only to **electric** charges
iii	Applies only to **point** charges
iv	It **diverges**: as $r \to 0$, $F \to \infty$
v	It **violates the principle of causality**, which says no physical information passes between two points faster than the speed of light. It is silent about the physical mechanism by which force is transferred between the charges. This is also called the principle of **locality** (all physical phenomena occur locally).

12. Electric field and electric field intensity

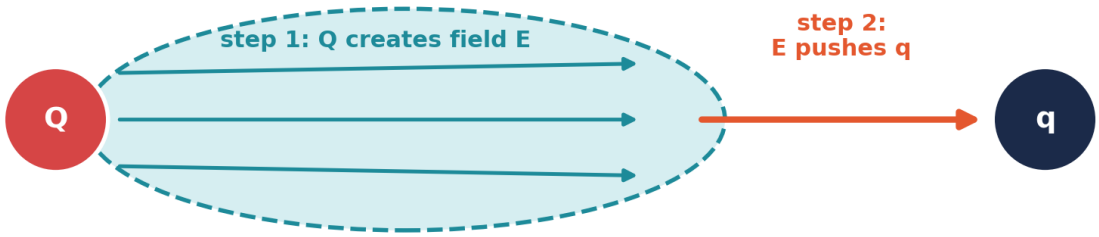
To make Coulomb’s law consistent with causality, **Michael Faraday** introduced the concept of the **electric field**. The force on a test charge +q due to a source charge Q is not instantaneous. It happens in two steps:

1. The source charge produces an electric field in its vicinity.
2. The electric field exerts a force on the test charge.

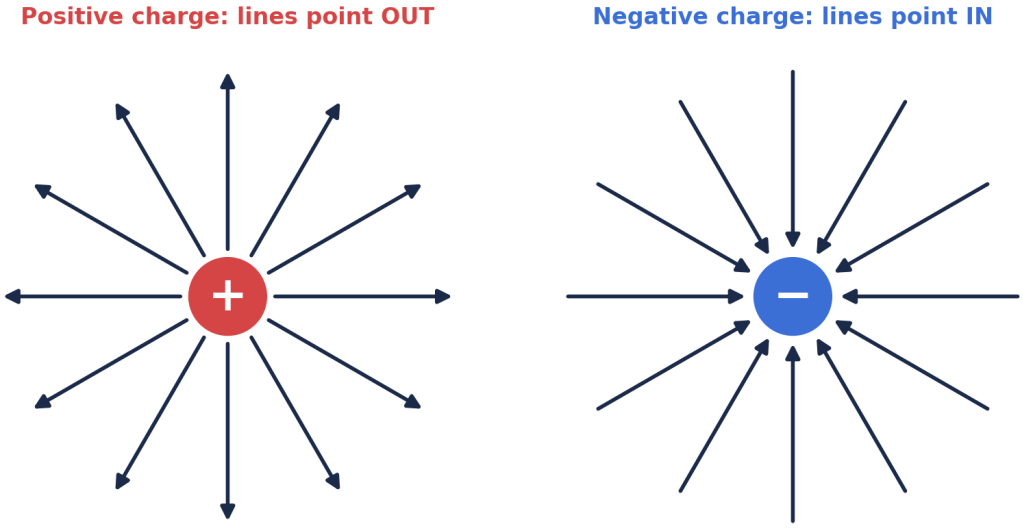
Coulomb alone: action at a distance



Faraday: the field does the work



Einstein introduced the concept of the gravitational field to overcome the same problem (E_g = F / m). Newton's law of gravitation, F = G $m_1 m_2$ / r^2, faced similar problems of non-locality.



The electric field is represented by straight field lines. For a positive source charge +Q they point radially outward.

Electric field intensity is the force per unit positive test charge:

$$\mathbf{E = F / q \Leftrightarrow F = q\,E}$$

- Field due to a point charge Q at distance r: $\mathbf{E = kQ / r^2}$.

- Field due to many source charges: divide the superposed force by q.

$$\mathbf{E = \Sigma\, k\, q_i\, \hat{r}_i / r_i^2}$$

Worked example

Q = 4 nC at r = 0.5 m.

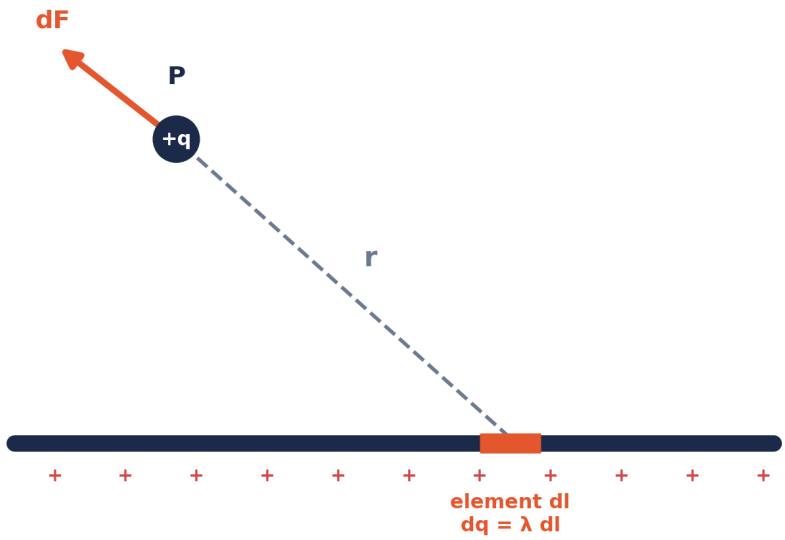
$$E = (9\times10^{9})(4\times10^{-9}) / (0.5)^2$$
$$E = 144\ N/C$$

REMEMBER

Coulomb's law gives the force directly, and the field is that force divided by the test charge. Knowing E, the force on any charge q is **F = qE**.

13. Continuous distribution of electric charge

Instead of forces between two stationary point charges, Coulomb's law can also be used to calculate the force from a **continuous distribution** of charge. Consider a line distribution on a wire, and find the force on a test charge +q at point P at distance r from a small element dl.



1. Line charge density λ = dq / dl, so **dq = λ dl**.
2. Force on +q from the element dl, by Coulomb's law:
3. Total force from the whole line: **integrate** over the length.

$$\mathbf{dF = k\, q\, dq\, \hat{r} / r^2 = k\, q\, \lambda\, dl\, \hat{r} / r^2}$$

$$\mathbf{F = k\, q \int (\lambda\, dl / r^2)\, \hat{r}}$$

Distribution	Density	Element of charge
Linear (along a line)	λ = dq / dl	dq = λ dl
Surface (over an area)	σ = dq / dA	dq = σ dA
Volume (throughout a volume)	ρ = dq / dV	dq = ρ dV

The same method works for all three, replacing dl by dA or dV. For the field, divide by q:

$$\mathbf{E = k \int (dq / r^2)\, \hat{r}}$$

BOARD QUESTION

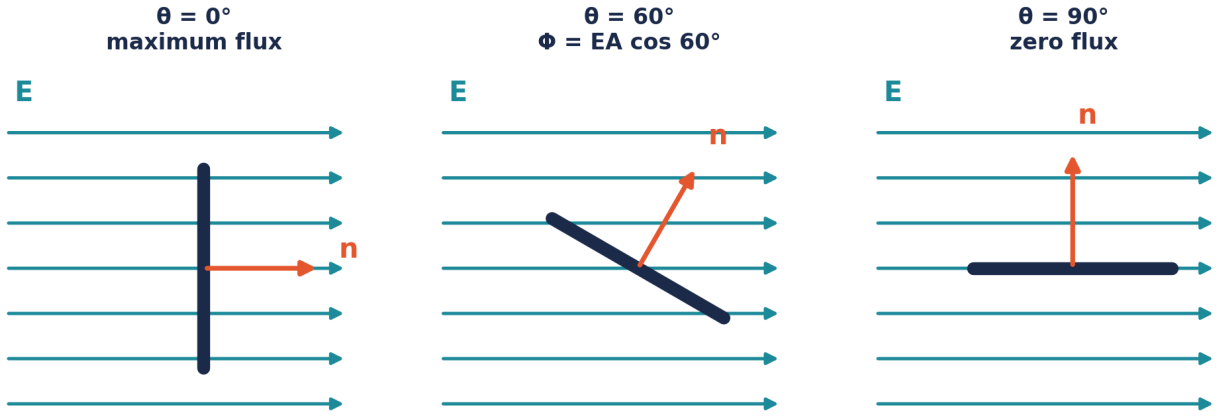
Board question: determine the force on a charge +q placed at a distance from (i) a line distribution, (ii) a surface (areal) distribution and (iii) a volume distribution.

14. Electric flux

Syllabus only for this topic.

- Electric flux measures **how much electric field passes through an area**. Pictured with field lines, it is proportional to the number of field lines crossing the surface.
- For a uniform field E through a flat area A, with θ the angle between E and the **normal** to the area:

$$\Phi_E = E \cdot A = E\,A\cos\theta,\ \Phi_E = \int E \cdot dA \text{ (general)}$$

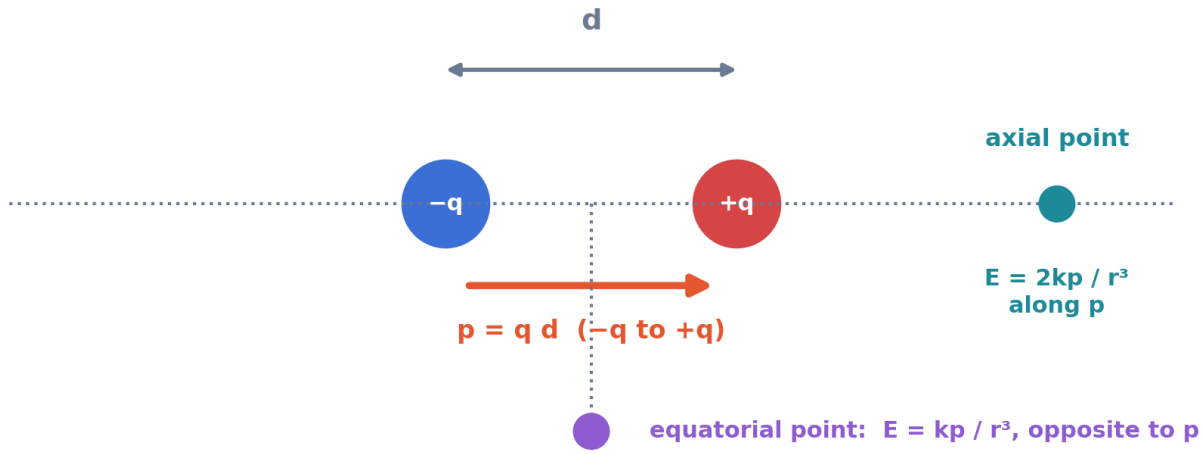


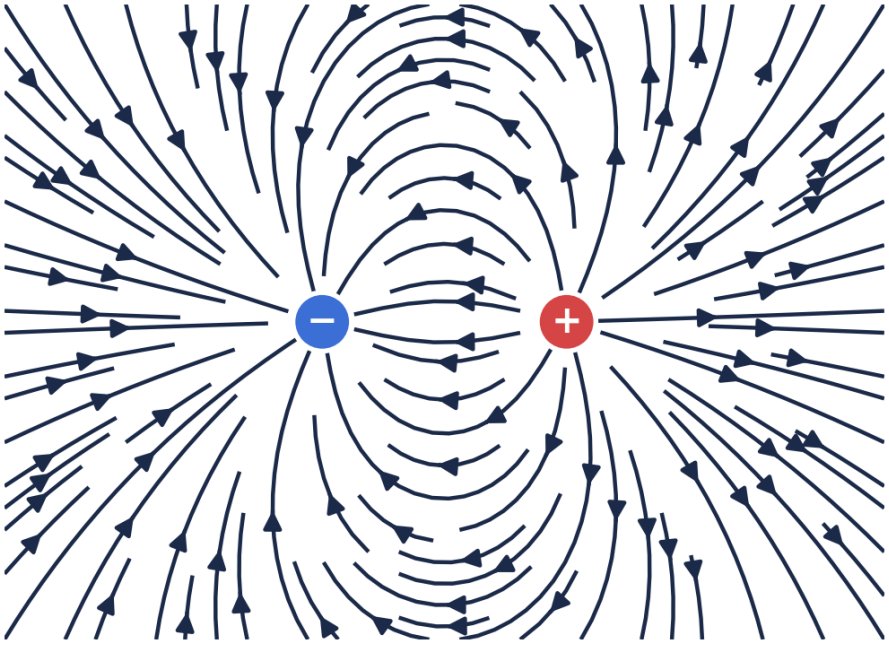
- Flux is **maximum** when the field is perpendicular to the surface (θ = 0), and **zero** when the field runs parallel to it (θ = 90°).
- Unit: **N m²/C**. It is a **scalar**.

15. Electric dipole

Syllabus only for this topic.

- An electric dipole is a pair of **equal and opposite charges** +q and −q separated by a small distance d.
- **Dipole moment p = q d**, a vector pointing from −q to +q.
- Field at a distance r much larger than d: on the **axis**, E = 2kp / r³, along p. On the **perpendicular bisector** (equatorial line), E = kp / r³, opposite to p.
- The dipole field falls as **1/r³**, faster than the 1/r² of a single point charge, because the two fields partly cancel.





Field lines of a dipole: they leave +q and end on −q.

$$E_{axial} = 2kp / r^3,\ E_{equatorial} = kp / r^3$$

Lab activities (not included in the mid-term)

Lab Activity 1: Ohm’s law

Data: V = 3, 6, 9, …, 24 V against I = 0.1, 0.2, …, 0.8 A (eight readings).

Question	Answer
Nature of the V vs I plot	A straight line through the origin: V and I are directly proportional
Average V and I	V = 108 / 8 = 13.5 V, I = 3.6 / 8 = 0.45 A
Slope	(18 − 6) / (0.6 − 0.2) = 30 V/A
Resistance R	30 Ω (check: 3 V / 0.1 A = 30 Ω)
Independent / dependent variable	Voltage V is independent; current I is dependent
Hypothesis	At constant temperature, current through a conductor is directly proportional to the voltage across it
Conclusion	V / I stays constant at 30 Ω, so the conductor obeys Ohm’s law, V = IR

WATCH THE AXES

The lecture convention puts **V on x and I on y**, so the slope is **1/R**. The lab answer key plots **V against I**, so the slope equals **R**. Say which axes you used and read the slope accordingly.

Lab Activity 2: errors and least count

- **Absolute error:** $\Delta x = |x_{measured} - x_{true}|$, with the same unit as the quantity.
- **Zero error:** the reading an instrument shows when it should read zero. Positive if above zero, negative if below. **True reading = observed reading − zero error.**
- **Fractional uncertainty** = $\Delta x / x$ (no unit). **Relative (percentage) uncertainty** = $(\Delta x / x) \times 100$ %. Example: 20.0 ± 0.1 cm gives 0.005, or 0.5 %.
- **Least count (LC):** the smallest value an instrument can measure accurately.

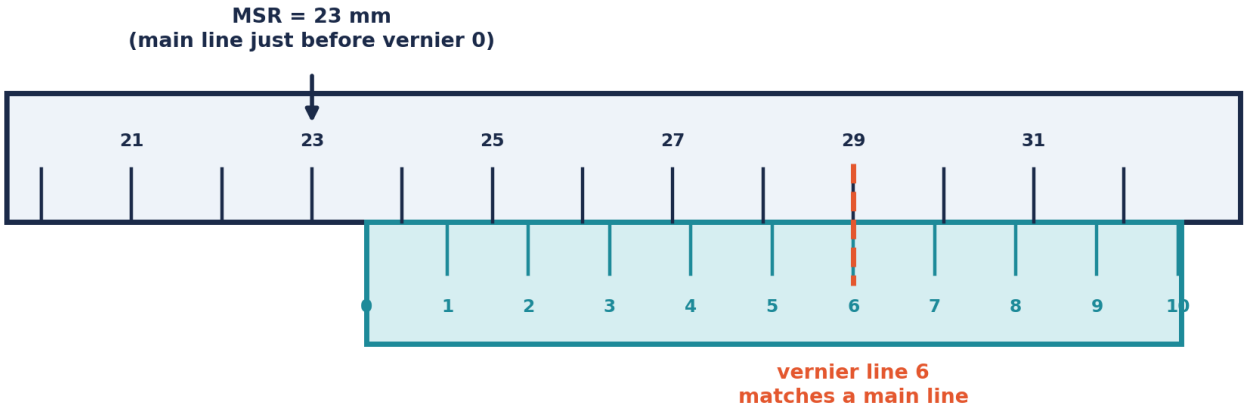
Instrument type	How to find the least count
Plain scale	Value of the smallest division
Vernier	1 main scale division ÷ number of vernier divisions
Micrometer	Pitch ÷ number of circular divisions

Instrument	Least count
Thermometer	1 °C
Vernier callipers	0.01 cm (0.1 mm)
Meter ruler	0.1 cm (1 mm)
Stop watch	0.01 s (digital)
Ammeter	0.02 A (example: 0–1 A, 50 divisions)
Voltmeter	0.1 V (example: 0–10 V, 100 divisions)
Galvanometer	1 scale division

Instrument	Least count
Top pan balance	0.01 g
Protractor	1°
Micrometer (screw gauge)	0.01 mm

Vernier calliper and screw gauge

	Vernier calliper	Screw gauge
Least count	1 MSD − 1 VSD = 1 mm − 0.9 mm = 0.1 mm (0.01 cm), because 10 VSD = 9 MSD	Pitch ÷ divisions = 0.5 mm ÷ 50 = 0.01 mm
Reading	MSR + VC × LC − ZE	LSR + CSR × LC − ZE
Positive zero error	+ n × LC	+ n × LC (n below 25)
Negative zero error	− (10 − n) × LC	− (50 − n) × LC (n above 25)
Correction	Subtract a positive ZE, add a negative ZE	Same



Reading = MSR + VC × LC = 23 + 6 × 0.1 = 23.6 mm = 2.36 cm

Example: vernier reading

```
MSR = 23 mm, vernier line 6 matches
observed = 23 + 6 × 0.1 = 23.6 mm = 2.36 cm
with ZE = +0.02 cm:  true = 2.34 cm
```

Example: screw gauge reading

```
LSR = 2.5 mm, CSR = 23
observed = 2.5 + 23 × 0.01 = 2.73 mm
with ZE = +0.03 mm:  true = 2.70 mm
```

GOOD PRACTICE

Check ZE first, find LC, avoid parallax, use the ratchet on the screw gauge, and take 3–5 readings and average them.

Quick revision

Formulas to write from memory

Topic	Formula
Ohm's law	$V = IR$; slope of I–V plot = $1/R$
Charge	$Q = It$
Galilean transformation	$x' = x - vt$, $u' = u - v$, $a' = a$, $F' = F$
Rotation of axes	$x' = x \cos\varphi + y \sin\varphi$, $y' = y \cos\varphi - x \sin\varphi$, and $\lvert r'\rvert = \lvert r\rvert$
Coulomb's law	$F = k\, q_1 q_2 / r^2$, $k = 9\times10^9$ N m²/C²
Electric field	$E = F / q$, $E = kQ / r^2$
Superposition	$F = \Sigma F_i$ and $E = \Sigma\, k\, q_i\, \hat{r}_i / r_i^2$
Continuous charge	$dq = \lambda\, dl$, $\sigma\, dA$ or $\rho\, dV$
Flux	$\Phi = EA \cos\theta$
Dipole	$p = qd$; $E_{axial} = 2kp/r^3$, $E_{equatorial} = kp/r^3$

Common mistakes

- ✗ Calling current a vector because it has a direction. It does not obey the vector addition law.
- ✗ Mixing up closure and "disclosure", or commutative and "communicative".
- ✗ Saying velocity is invariant under a Galilean transformation. **Acceleration and force** are invariant; velocity is not.

- ✗ Forgetting to subtract a positive zero error (and to add a negative one).
- ✗ Treating Coulomb’s law as valid for moving charges or at r = 0.
- ✗ Dropping the vector sum in superposition and adding magnitudes instead.
- ✗ Putting the dependent variable on the x-axis.

Self-test

1. Why is current a scalar?
2. Show that F = F′ under a Galilean transformation, in four steps.
3. What is the charge of a proton and of a neutron in terms of quark charges?
4. Force between 2 µC and 3 µC charges 10 cm apart?
5. Electric field 0.5 m from a 4 nC point charge?
6. Name the five limitations of Coulomb’s law.
7. Vernier with LC = 0.01 cm: MSR = 2.3 cm, VC = 6, ZE = +0.02 cm. True reading?
8. Name the three quantum revolutions and one application of each.

Self-test answers

Try the questions first. No peeking!

1. Currents add as plain numbers at a junction ($I = I_1 + I_2$), not by the vector triangle law, so current is a scalar even though it has a direction along the wire.
2. $x = vt + x'$, then $u = v + u'$, then $a = a'$ (v constant), then $F = ma = ma' = F'$.
3. **Proton** uud = 2/3 + 2/3 − 1/3 = +1. **Neutron** udd = 2/3 − 1/3 − 1/3 = 0.
4. **5.4 N** (repulsive, since both charges are positive).
5. **144 N/C**.
6. Stationary charges only; electric charges only; point charges only; diverges as $r \to 0$; violates causality / locality.
7. Observed = 2.3 + 6 × 0.01 = 2.36 cm. True = 2.36 − 0.02 = **2.34 cm**.
8. **First:** atomic (energy level) theory, for example lasers. **Second:** energy band theory, for example transistors and chips. **Third:** quantum computing.